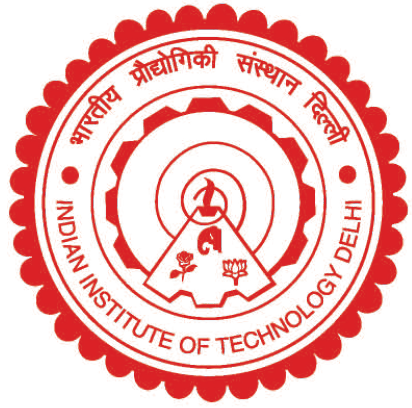
Study of Magnetic Fluids Showing Negative Viscosity Behavior Using Lattice Boltzmann Method (LBM)



Department of Chemical Engineering
Indian Institute of Technology, Delhi

Submitted by: Snigdhadyut Dash (2022CH71482)
Supervisor: Prof. Jayati Sarkar

Mid-Sem Presentation · 24 September 2026

Outline

1. Introduction & Motivation
2. Magnetic-fluid specifications
3. Vector potential — derivation & justification
4. LBM governing equations & framework
5. Solver validation — four tests
6. Purely magnetically driven flow
7. Future work, references & acknowledgments

Introduction & Motivation

- **Problem Statement:** The microscale coupling between magnetic torque and viscous torque governs how a ferrofluid responds to a field, but it is difficult to predict at the macroscale. Under a suitable oscillating field the effective viscosity can fall, and at strong coupling it becomes negative.
- **Motivation:** A tunable, even negative, apparent viscosity enables valveless micro-pumping, adaptive dampers, and field-guided transport for targeted drug delivery.
- **Objectives:**
  1. Establish the 2D/3D torque balances that govern the spin–flow coupling.
  2. Build a robust phase-field MRT Lattice Boltzmann solver for two-fluid magnetic flow.
  3. Validate the solver against analytical limits and the Bacri–Perzynski (1995) benchmark.

Magnetic-Fluid Specifications

The benchmark ferrofluid of Bacri, Perzynski & Shliomis (PRL 75, 1995)

Physical properties

- Cobalt-ferrite nanoparticles (Massart synthesis)
- Grain magnetization $M_s = 400$ G
- Particle size ~ 10 nm; anisotropy $K = 2 \times 10^6$ erg/cm^3 (rigid dipole)
- Carrier: water; volume fraction $\phi = 20\%$
- Brownian time $\tau_B \approx 1.6$ ms
- Newtonian: $\eta = 77$ cP (zero field) $\rightarrow 220$ cP (static 2000 Oe)

Governing balance

$$6\eta\phi(\boldsymbol{\omega}_p - \boldsymbol{\Omega}) = \mathbf{M} \times \mathbf{H}$$

Rotational-viscosity change

$$\Delta\eta = \tfrac{3}{2}\,\eta\phi\,\frac{\Omega - \omega_p}{\Omega}$$

Control parameters

$$\xi_0 = \frac{M_s V H}{k_B T}, \qquad \omega\tau_B$$

Benchmark result: field frequency alone tunes the viscosity from 220 down to 50 cP (over 0–700 Hz), i.e. $\Delta\eta$ swings from $+143$ to -19 cP. The marginal (zero-viscosity) state is predicted at $\xi_c = 126$, $\omega_c\tau_B = 20$.

The Vector-Potential Derivation

Quasi-static Maxwell equations, at every instant:

$$\nabla \times \mathbf{H} = 0, \qquad \nabla \cdot \mathbf{B} = 0, \qquad \mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}).$$

In 2D, a single scalar A_z satisfies the divergence constraint automatically:

$$\mathbf{B} = \nabla \times (A_z \hat{\mathbf{z}}) = (\partial_y A_z,\ -\partial_x A_z) \implies \nabla \cdot \mathbf{B} \equiv 0.$$

Imposing $\nabla \times \mathbf{H} = 0$ gives the governing equation and field recovery:

$$\nabla^2 A_z = -\mu_0 (\nabla \times \mathbf{M})_z, \qquad \mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}.$$

	Scalar ψ (old)	**Vector A_z (current)**
Poisson equation	$\nabla^2 \psi = \nabla \cdot \mathbf{M}$	$\nabla^2 A_z = -\mu_0 (\nabla \times \mathbf{M})_z$
Source	divergence of $\mathbf{M}$	curl of $\mathbf{M}$
$\nabla \cdot \mathbf{B} = 0$	imposed	automatic
Mean field along x	lost (topological)	carried by wall boundary condition

The same Poisson solver is reused; only the source term and the recovery formula change.

Justification: Vector versus Scalar Potential

Why the scalar potential fails on a periodic channel

- **Scalar choice:** one value per node, $\mathbf{H} = -\nabla\psi$, solving

$$\nabla^2\psi = \nabla \cdot \mathbf{M}.$$

  It is cheap, but it cannot carry a field along the channel.

- **Why it fails:** the channel repeats in x, so ψ returns to the same value each period. Its average slope along x is then zero,

$$\langle H_x \rangle = -\frac{1}{L}\int_0^L \partial_x\psi \, dx = -\frac{\psi(L,y) - \psi(0,y)}{L} = 0,$$

  and the applied field disappears. This is a property of the closed loop, not of static versus oscillating fields.

- **The fix:** build the field from a vector potential A_z,

$$\mathbf{B} = \nabla \times (A_z\hat{\mathbf{z}}), \qquad H_x = \tfrac{1}{\mu_0}\,\partial_y A_z.$$

  A steady field along x is stored as a change in y (the walled direction), where no loop closes, so the field is kept.

LBM Governing Equations (D2Q9)

Interface — conservative Allen–Cahn (g-lattice):

$$g_k^{eq} = w_k \phi \left[1 + \frac{\mathbf{c}_k \cdot \mathbf{u}}{c_s^2}\right] + w_k \frac{M_\phi}{c_s^2} \frac{4}{D} \phi(1-\phi)\, \mathbf{c}_k \cdot \hat{\mathbf{n}}, \qquad \phi = \sum_k g_k.$$

Hydrodynamics — pressure-based MRT (f-lattice):

$$f_k^{eq} = w_k\, p + \rho c_s^2 (\Gamma_k - w_k), \qquad \mathbf{F}_{tot} = \mathbf{F}_s + \mathbf{F}_m + \mathbf{F}_b.$$

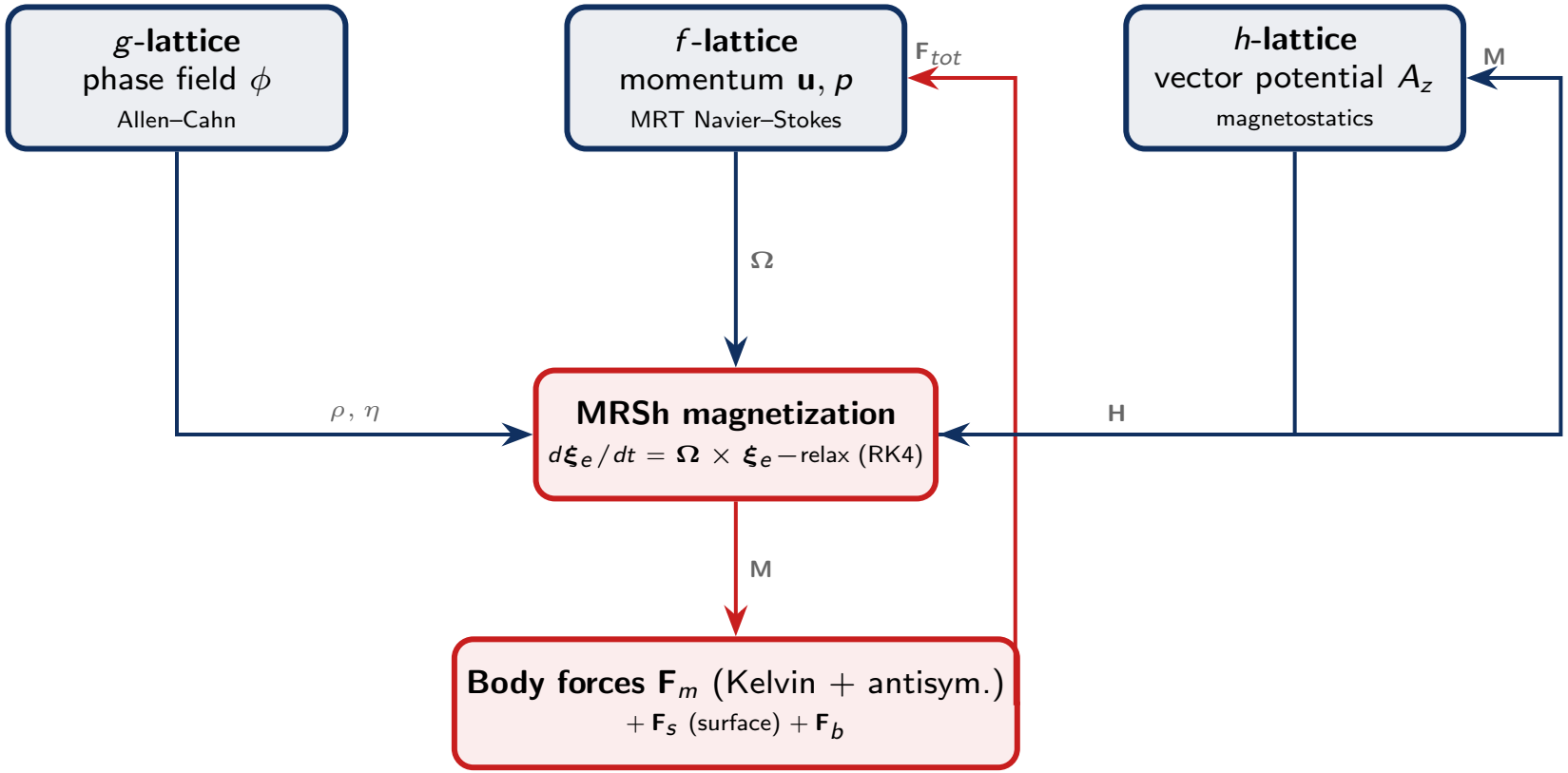
Magnetization dynamics — MRSh effective-field model:

$$\mathbf{M} = \phi M_s L(\xi_e)\hat{\mathbf{e}}, \qquad \frac{d\boldsymbol{\xi}_e}{dt} = \boldsymbol{\Omega} \times \boldsymbol{\xi}_e - \frac{1}{\tau_B}\frac{L}{L'}\left(1 - \frac{\xi_\parallel}{\xi_e}\right)\hat{\mathbf{e}} + \frac{1}{2\tau_B}\left(\frac{\xi_e}{L} - 1\right)\boldsymbol{\xi}_\perp.$$

Magnetostatics (h-lattice) and **body force**:

$$\nabla^2 A_z = -\mu_0 (\nabla \times \mathbf{M})_z, \quad \mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}, \quad \mathbf{F}_m = \mu_0 (\mathbf{M}\cdot\nabla)\mathbf{H} + \tfrac{1}{2}\nabla \times (\mu_0 \mathbf{M} \times \mathbf{H}).$$

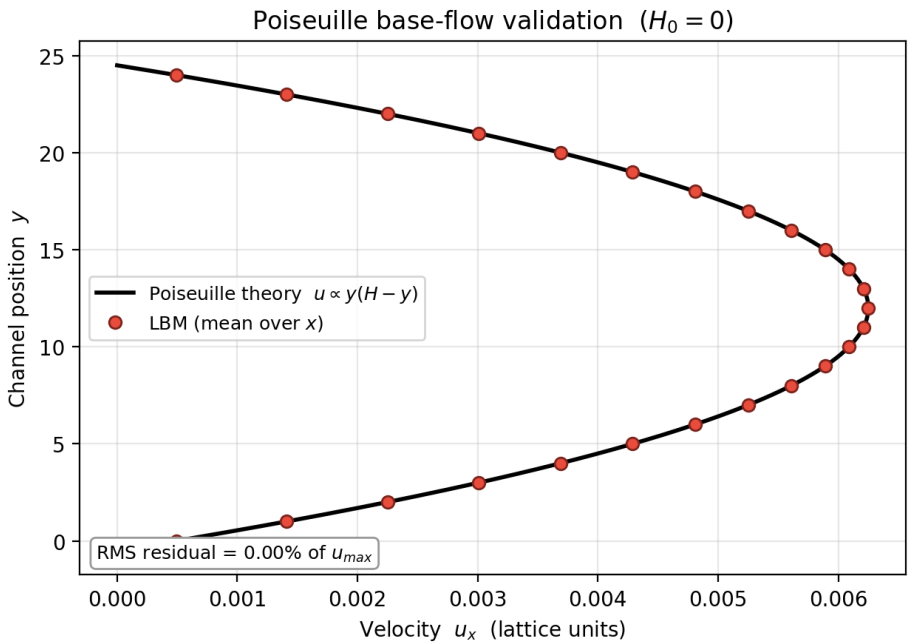
The LBM Framework — Three Coupled Lattices



The flow tilts the magnetization; the magnetization sets the body force; the force drives the flow. The three lattices are advanced together every timestep.

Base-Flow Accuracy — Poiseuille Profile

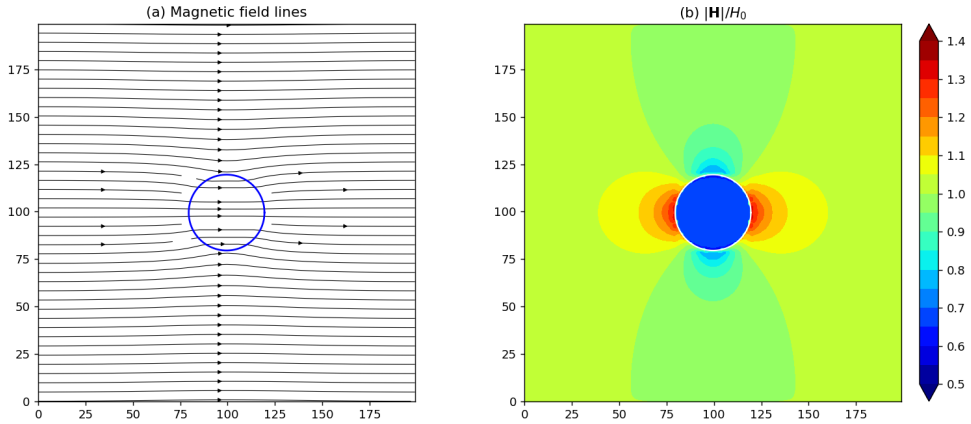
The most basic check: the hydrodynamic solver before any magnetic coupling

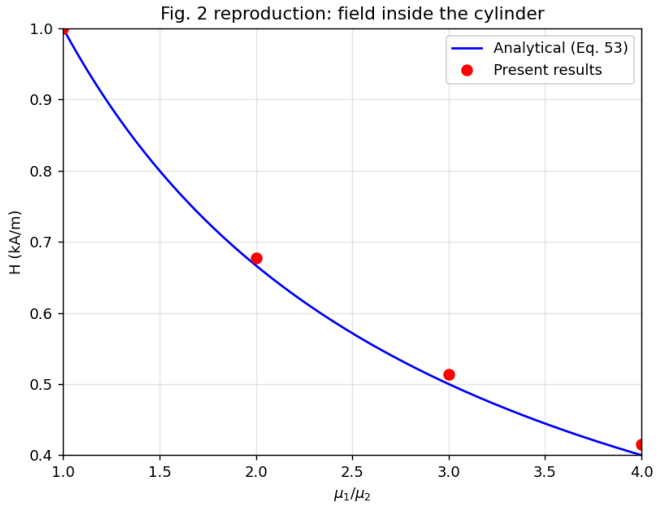


- **Setup:** a pressure-driven channel with no applied field; the LBM profile is averaged over x and compared to the analytic parabola.
- **Result:** the profile lands on the theory with a root-mean-square residual near zero, confirming a mass- and momentum-conserving solver.

Validation 1 — Bending of Magnetic Field Lines

Pure magnetostatics: a magnetizable cylinder in a uniform field, flow frozen

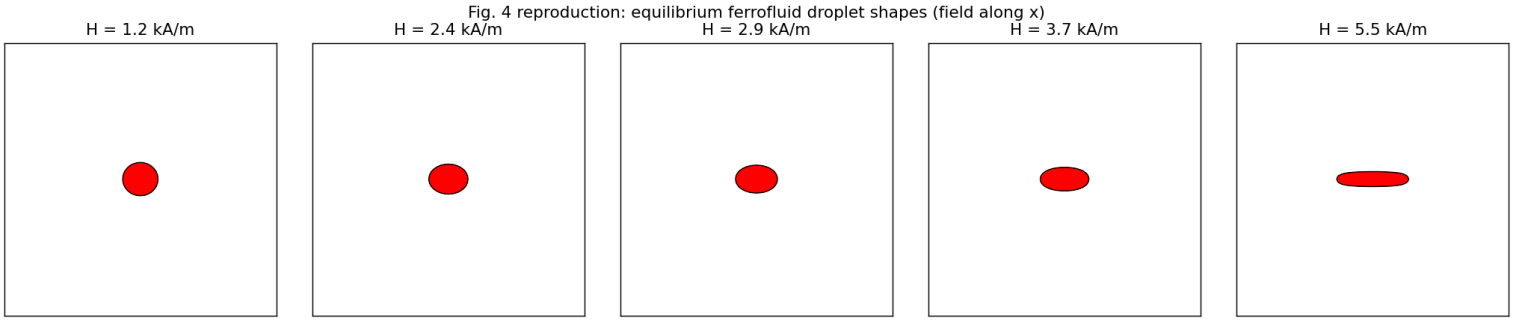


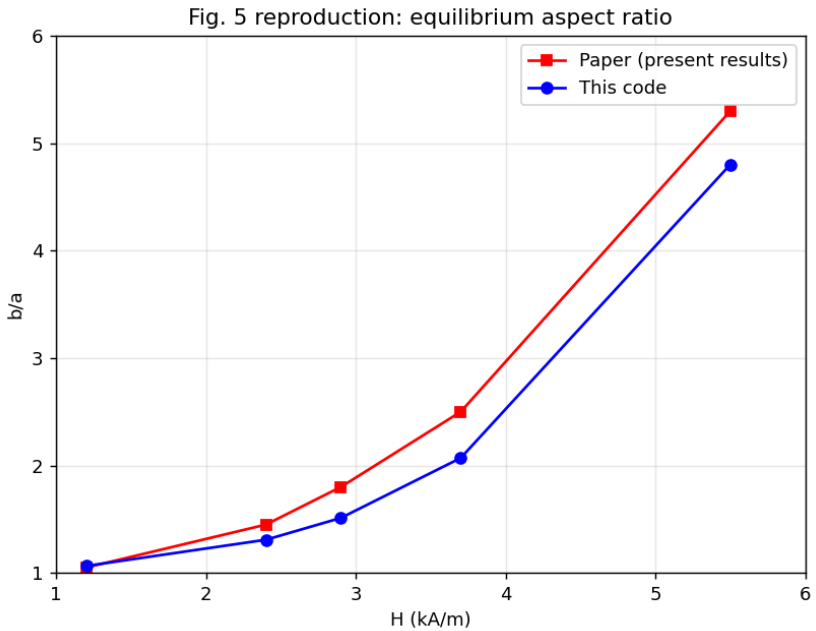


- **Setup:** a circular body of higher permeability in a uniform applied field, with the flow solver switched off to isolate the *h*-lattice.
- **Result:** the interior field matches the analytic value $H_{\text{in}}/H_0 = 2/(\mu_1/\mu_2 + 1)$ to within 0–3.8% across the permeability sweep; the residual is finite-domain far-field truncation, not solver error.
- **Observation:** the more permeable body draws field lines into itself, giving a uniform interior field whose strength is set only by the permeability ratio.

Validation 2 — Ferrofluid Droplet Elongation

Coupled field, interface and flow: a droplet stretches into an ellipse

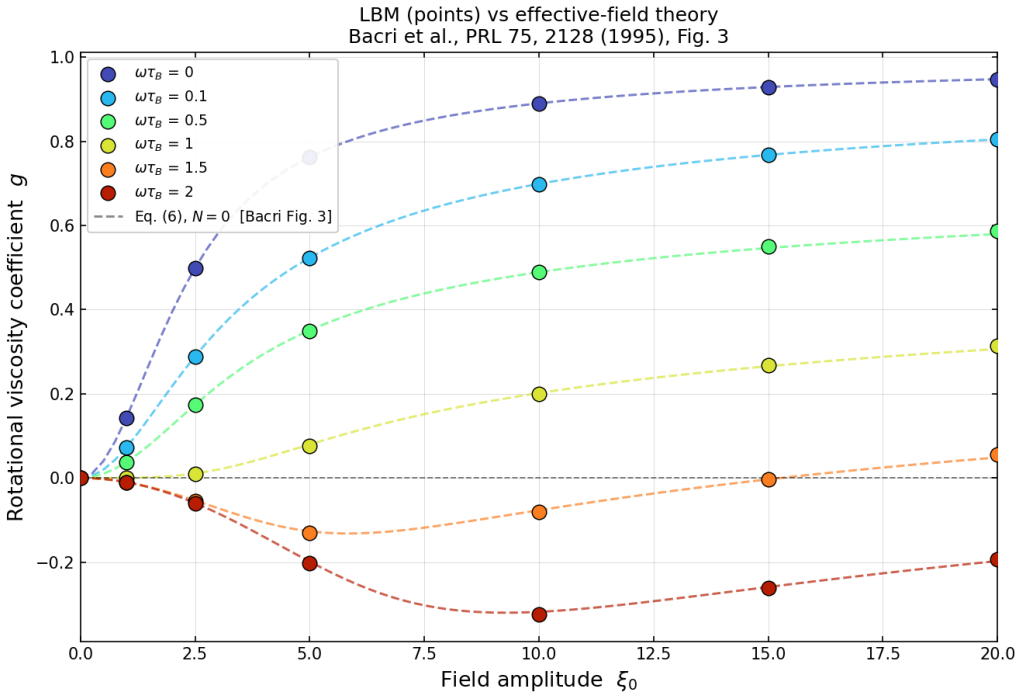




- **Setup:** magnetic stress elongates the droplet along the field while surface tension resists; the aspect ratio b/a is governed by the magnetic Bond number $\mathrm{Bo}_m = \chi_0^2 H^2 R/\sigma$.
- **Result:** the solver reproduces the reference convex $b/a(H)$ curve across the field range.
- **Observation:** at the two highest fields the elongated droplet wraps in the periodic-x box; these points recover once the domain is enlarged.

Validation 3 — Bacri Figure 3 (central target)

Poiseuille flow under an oscillating field: the negative-viscosity effect



- **Setup:** a body-force-driven channel flow under a linearly oscillating field; the rotational-viscosity coefficient $g(\xi_0, \omega\tau_B)$ is extracted from the converged profile.
- **Result:** the LBM points lie on the effective-field theory curves, with g turning negative at large ξ_0 and high $\omega\tau_B$.
- **Accuracy:** $\sim 0.3\%$ in static fields; $\sim 1.6\%$ up to $\xi_0 = 10$, rising to $\sim 4.5\%$ at $\xi_0 = 20$.

Purely Magnetically Driven Flow — Setup

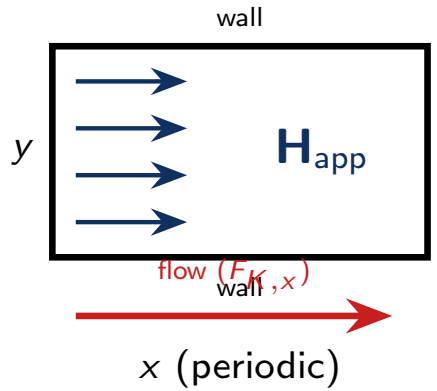
Field formulation, driving mechanism, and boundary conditions

- **Formulation:** a harmonic, centreline-symmetric applied potential,

$$A_{\text{app}} = \mu_0(H_d\, Y + k\, XY),$$

$$\mathbf{H}_{\text{app}} = (H_d + kX,\ -kY).$$

- **Mechanism:** the streamwise Kelvin force $F_{K,x} \approx \mu_0 M(H_d)\, k$ replaces the pressure gradient. A non-zero H_d is required, since a pure-gradient field has a null at the centre and gives no net drive.
- **Values:** $H_d = 1.73$, $k = 1.21 \times 10^{-5}$ (matched to $G = 2 \times 10^{-7}$).



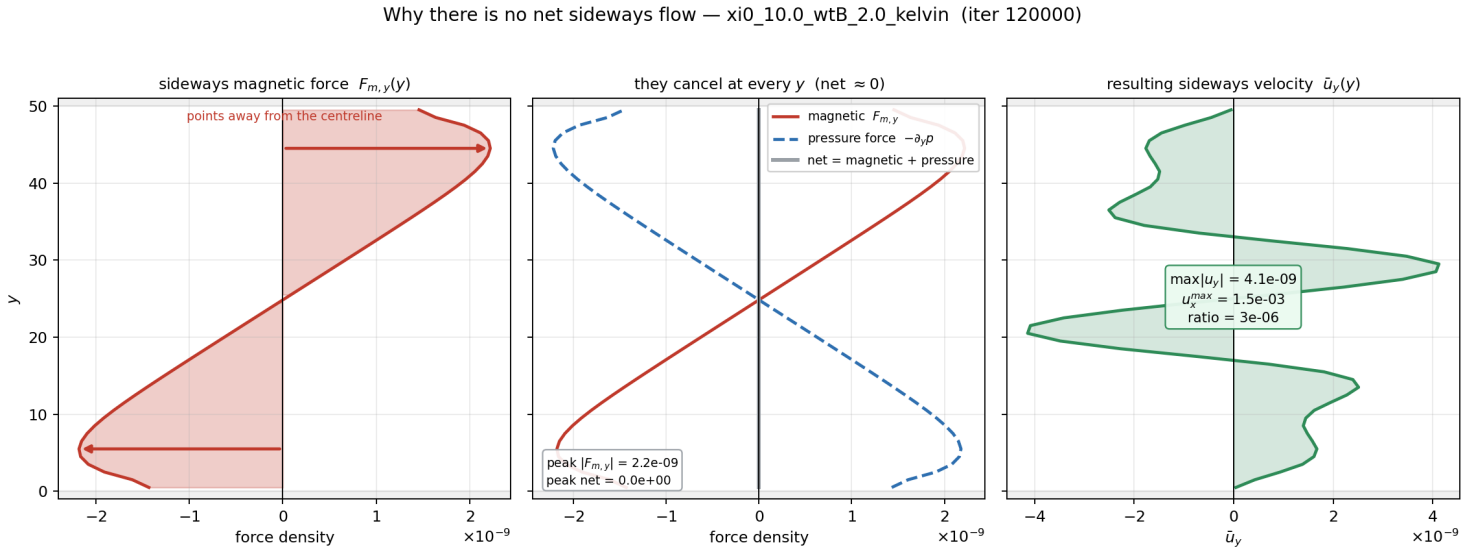
Boundary conditions

- Walls (f, g): no-slip via halfway bounce-back.
- Walls (h): Neumann on A_z, $\partial A_z/\partial n = \mu_0(\mathbf{H}_{\text{app}} + \mathbf{M}_w) \cdot \hat{\mathbf{t}}$.
- x: periodic; jump-periodic seam on A_z (A_{app} is not periodic).
- Single-phase runs ($\phi \equiv 1$).

The Transverse Force Balance (y-direction)

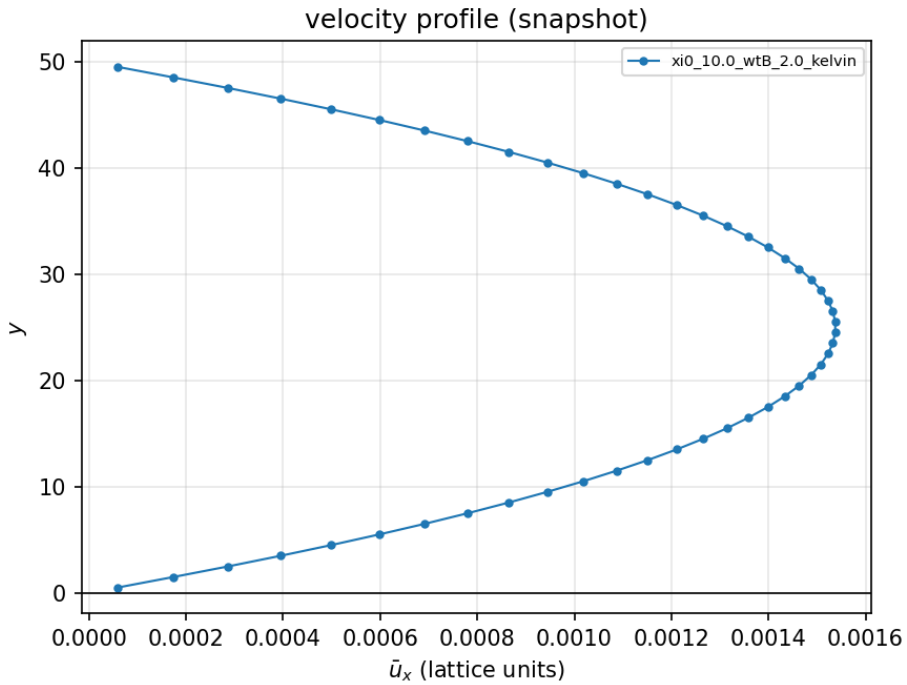
Confirming the setup stays unidirectional and fully developed

Purpose: verify the assumption behind the boundary conditions — that the transverse force does not disturb the channel flow.



- **Observation:** at every height the transverse magnetic force $F_{m,y}$ (red) is cancelled by the pressure force (blue); the net transverse velocity is $|u_y| \sim 10^{-9}$ against $u_x^{\max} \sim 10^{-3}$.
- **Mechanism:** $F_{m,y}$ depends on y alone, so it is a pure gradient that pressure absorbs exactly. Only the mean streamwise force survives, because periodicity prevents pressure from developing a mean slope along x.

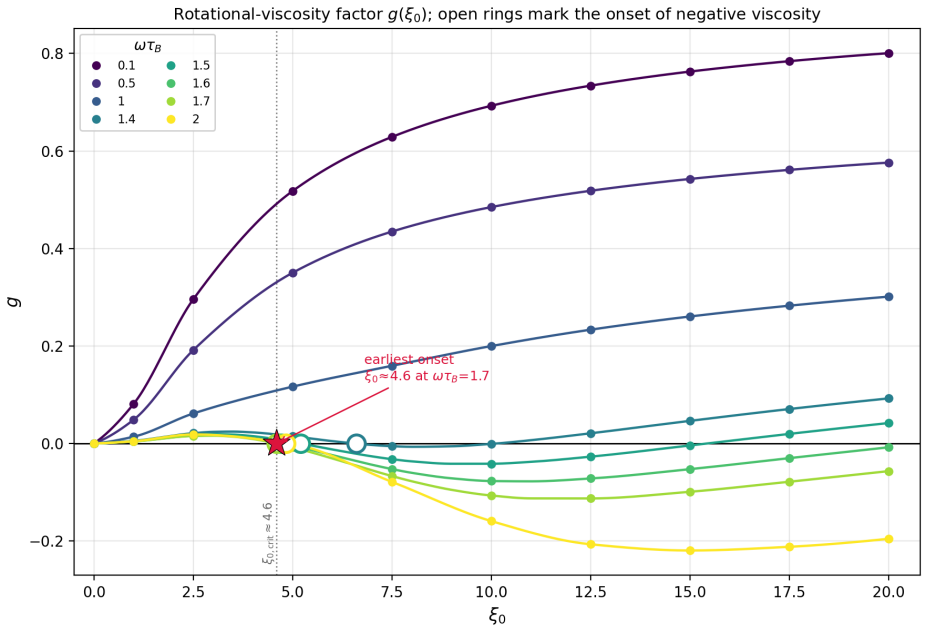
Velocity Profile under Purely Magnetic Drive



- **Result:** the field alone produces a clean parabolic profile with $u_x^{\max} \approx 1.5 \times 10^{-3}$.
- **Observation:** the transverse force leaves no imprint on the profile; no secondary circulation appears.
- **Implication:** a spatially graded magnetic field reproduces textbook Poiseuille flow, demonstrating a valveless magnetic pump.

Negative Viscosity under Magnetic Drive

The Bacri Figure 3 equivalent for the purely Kelvin-driven flow



- **Persistence:** the effect survives with *zero* external pressure gradient; the same field both pumps and modulates the flow.
- **Threshold:** viscosity stays positive until $\xi_0 \approx 4.6$, beyond which g turns negative.
- **Optimal drive:** onset is reached at the smallest field near $\omega\tau_B \approx 1.7$.

Future Work

- **Coupled oscillating shear and field:** the change in apparent viscosity when the fluid is subjected to both an oscillating shear and an oscillatory magnetic field at the same frequency ω.
- **Full two-phase Kelvin-driven flow:** couple the water/ferrofluid interface to the purely magnetically driven configuration.
- **Thermal coupling:** add the energy equation for temperature-dependent magnetization and thermomagnetic convection.
- **Antisymmetric-field pumping:** directional flow generated by specially tailored antisymmetric oscillating fields.
- **Three-dimensional extension:** generalize the vector-potential formulation beyond the planar channel.

References & Acknowledgments

References

1. Shliomis, M. I. (2002). Effective viscosity of magnetic fluids. *Physical Review E*, **64**(6).
2. Bacri, J.-C., & Perzynski, R. (1995). "Negative-Viscosity" Effect in a Magnetic Fluid. *Physical Review Letters*, **75**(11).
3. Hu, Y., & Li, D. (2018). Phase-field-based lattice Boltzmann model for multiphase ferrofluid flows. *Physical Review E*, **98**, 033301.

Acknowledgments

I sincerely thank my supervisor, Prof. Jayati Sarkar, for her guidance throughout this work, and the Department of Chemical Engineering, IIT Delhi, for its support.



Thank You